

ECON1050 Tutorial 12

Matrix Algebra III

Week 13 - Semester 1 2026

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Today's Plan

EXam: ① review all the lecture
② try all past exams: AI
③ Summary

- ▶ Recap: key ideas from Lecture 12
- ▶ Tutorial questions: Q1–Q5

Start at 8:05



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Recap: Cramer's Rule

Core idea

For a system

$$\underline{A} \cdot \underline{x} = b$$

$$\mathbf{A}\mathbf{x} = \mathbf{b},$$

Cramer's Rule uses determinants to solve for one variable at a time.

Formula

If $|\mathbf{A}| \neq 0$, then

$$\underline{x_k} = \frac{|\mathbf{A}_k|}{|\mathbf{A}|},$$

where $\mathbf{A}_k$ is formed by replacing the k th column of $\mathbf{A}$ with the constants vector $\mathbf{b}$.

When useful

Cramer's Rule is useful when we only need one endogenous variable.

Recap: Cramer's Rule for a 3×3 system

System

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

$\underset{\text{A}}{\quad} \quad \underset{\text{x}}{\quad} \quad \underset{\text{b}}{\quad}$

Replacement matrices

$$x_1 = \frac{|A_1|}{|A|}, \quad x_2 = \frac{|A_2|}{|A|}, \quad x_3 = \frac{|A_3|}{|A|}.$$

Each A_k replaces column k of A with b .

$$x_1 = \frac{|A_1|}{|A|} \quad A_1 = \begin{pmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{pmatrix}$$

x_2
 x_3

Recap: Minors and cofactors

Minor

For a square matrix $\mathbf{A}$, the minor M_{ij} is the determinant of the submatrix obtained by deleting row i and column j .

Cofactor

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

A cofactor exists for every element of a square matrix.

Matrix of cofactors

The matrix of cofactors is denoted by $\mathbf{C}^+$.

$$2 \times 2 \quad \mathbf{C}^+ = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}$$

Recap: The adjoint method

Adjoint / adjugate

The transpose of the matrix of cofactors is called the adjoint:

$$\text{adj}(\mathbf{A}) = (\mathbf{C}^+)'.$$

$$\mathbf{C}^+ = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix}$$

$$(\mathbf{C}^+)' = \begin{pmatrix} c_{11} & c_{21} \\ c_{12} & c_{22} \end{pmatrix}$$

Inverse using the adjoint

If $|\mathbf{A}| \neq 0$, then

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj}(\mathbf{A}).$$

Procedure

Find $|\mathbf{A}|$, find all cofactors, transpose the cofactor matrix, then divide by $|\mathbf{A}|$.

Recap: Solving systems using the inverse

Inverse method

If

$$Ax = b$$

and A^{-1} exists, then

$$\underline{x} = A^{-1}b.$$

$$\begin{aligned} Ax &= b \\ A^{-1}Ax &= A^{-1}b \\ Ix &= A^{-1}b \\ x &= A^{-1}b \end{aligned}$$

Non-singularity condition

The inverse exists if and only if

$$|A| \neq 0.$$

Tutorial Q1

Question 1

Solve for x_2 in the following equation system using Cramer's rule.

$$Ax = b \Leftrightarrow \begin{cases} 2x_1 + 3x_2 - x_3 = -1 \\ 5x_1 + x_2 + x_3 = -2 \\ -x_1 - x_2 + 4x_3 = 8 \end{cases}$$
$$\begin{pmatrix} 2 & 3 & -1 \\ 5 & 1 & 1 \\ -1 & -1 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ 8 \end{pmatrix}$$

$$x_2 = \frac{|A_2|}{|A|} = \frac{-49}{-49} = 1$$

$$|A| = \begin{vmatrix} 2 & 3 & -1 \\ 5 & 1 & 1 \\ -1 & -1 & 4 \end{vmatrix} = 2 \times 1 \times 4 + 3 \times 1 \times (-1) + (-1) \times 5 \times (-1) - 1 \times (-1) \times (-1) - 5 \times 3 \times 4 - 2 \times 1 \times (-1) = -49$$

$$A_2 = \begin{pmatrix} 2 & -1 & -1 \\ 5 & -2 & 1 \\ -1 & 8 & 4 \end{pmatrix} \Rightarrow |A_2| = 2 \times \begin{vmatrix} -2 & 1 \\ 8 & 4 \end{vmatrix} - (-1) \times \begin{vmatrix} 5 & 1 \\ -1 & 4 \end{vmatrix} + (-1) \times \begin{vmatrix} 5 & -2 \\ -1 & 8 \end{vmatrix}$$
$$= 2 \times (-8 - 8) - (-1) \times (5 \times 4 + 1) + (-1) \times (5 \times 8 - 2)$$
$$= -49$$



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Tutorial Q1: Cramer's Rule

Knowledge used

- Use Cramer's Rule:

$$x_k = \frac{|\mathbf{A}_k|}{|\mathbf{A}|}.$$

- $\mathbf{A}_k$ replaces the k th column of $\mathbf{A}$ with $\mathbf{b}$.

Solution

$$|\mathbf{A}| = \begin{vmatrix} 2 & 3 & -1 \\ 5 & 1 & 1 \\ -1 & -1 & 4 \end{vmatrix} = -49.$$

$$|\mathbf{A}_1| = 49, \quad |\mathbf{A}_2| = -49, \quad |\mathbf{A}_3| = -98.$$

Therefore,

$$x_1 = \frac{49}{-49} = -1, \quad x_2 = \frac{-49}{-49} = 1, \quad x_3 = \frac{-98}{-49} = 2.$$

$$(x_1, x_2, x_3) = (-1, 1, 2).$$

Tutorial Q2

Question 2

Consider the following macroeconomic model:

7 min

$$Y = C + \bar{I} + \bar{G}$$

$$C = \alpha + \beta(Y - T) \quad (\alpha > 0, 0 < \beta < 1)$$

$$T = \gamma + \delta Y \quad (\gamma > 0, 0 < \delta < 1)$$

where

Y = national income

C = consumption

$\bar{I}$ = investment, exogenously determined

$\bar{G}$ = government spending, exogenously determined

Using Cramer's rule, determine the equilibrium value T^* .



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Question 2

Consider the following macroeconomic model:

$$Y = C + \bar{I} + \bar{G}$$

$$C = \alpha + \beta(Y - T) \quad (\alpha > 0, 0 < \beta < 1)$$

$$T = \gamma + \delta Y \quad (\gamma > 0, 0 < \delta < 1)$$

where

Y = national income

C = consumption

$\bar{I}$ = investment, exogenously determined

$\bar{G}$ = government spending, exogenously determined

Using Cramer's rule, determine the equilibrium value T^* .

Endogenous variable: Y, C, T

exogenous variable: $\bar{I}, \bar{G}, d, \beta, \gamma, \delta$

$$Y = C + \bar{I} + \bar{G} \Rightarrow Y - C = \bar{I} + \bar{G}$$

$$C = d + \beta(Y - T) \Rightarrow -\beta Y + C + \beta T = d$$

$$T = \gamma + \delta Y \Rightarrow -\delta Y + T = \gamma$$

$$\begin{pmatrix} 1 & -1 & 0 \\ -\beta & 1 & \beta \\ -\delta & 0 & 1 \end{pmatrix} \begin{pmatrix} Y \\ C \\ T \end{pmatrix} = \begin{pmatrix} \bar{I} + \bar{G} \\ d \\ \gamma \end{pmatrix}$$

Cramer's rule: $T = \frac{|A_3|}{|A|} = \frac{\begin{vmatrix} 1 & -1 & \bar{I} + \bar{G} \\ -\beta & 1 & d \\ -\delta & 0 & \gamma \end{vmatrix}}{\begin{vmatrix} 1 & -1 & 0 \\ -\beta & 1 & \beta \\ -\delta & 0 & 1 \end{vmatrix}}$

$$= \frac{\gamma + d\delta + 0 + \delta(\bar{I} + \bar{G}) - \beta\gamma}{1 + \beta\delta + 0 - 0 - \beta - 0}$$

$$= \frac{\gamma + d\delta + \delta(\bar{I} + \bar{G}) - \beta\gamma}{1 + \beta\delta - \beta}$$



Tutorial Q2: Matrix form

Knowledge used

- ▶ Rearrange equations so that endogenous variables are on the left.
- ▶ Write the system as $\mathbf{Ax} = \mathbf{b}$.

Solution

Rearrange:

$$Y - C = \bar{I} + \bar{G}, \quad -\beta Y + C + \beta T = \alpha, \quad -\delta Y + T = \gamma.$$

Hence

$$\begin{bmatrix} 1 & -1 & 0 \\ -\beta & 1 & \beta \\ -\delta & 0 & 1 \end{bmatrix} \begin{bmatrix} Y \\ C \\ T \end{bmatrix} = \begin{bmatrix} \bar{I} + \bar{G} \\ \alpha \\ \gamma \end{bmatrix}.$$

Tutorial Q2: Solving for tax revenue

Knowledge used

- Use Cramer's Rule for the variable T .
- Replace the T column of $\mathbf{A}$ with the constants vector.

Solution

$$|\mathbf{A}| = \begin{vmatrix} 1 & -1 & 0 \\ -\beta & 1 & \beta \\ -\delta & 0 & 1 \end{vmatrix} = 1 - \beta(1 - \delta).$$

$$|\mathbf{A}_T| = \begin{vmatrix} 1 & -1 & \bar{l} + \bar{G} \\ -\beta & 1 & \alpha \\ -\delta & 0 & \gamma \end{vmatrix} = \delta(\alpha + \bar{l} + \bar{G}) + \gamma(1 - \beta).$$

Thus

$$T^* = \frac{\delta(\alpha + \bar{l} + \bar{G}) + \gamma(1 - \beta)}{1 - \beta(1 - \delta)}.$$

Tutorial Q3

Question 3

Find the inverse using the adjoint method:

$$(1) \begin{bmatrix} 5 & 2 \\ 1 & -1 \end{bmatrix}, \quad (2) \begin{bmatrix} 11 & -6 \\ 18 & -10 \end{bmatrix} :$$

$$(3) \begin{bmatrix} 1 & 0 & -2 \\ 4 & -2 & 1 \\ 1 & 2 & -10 \end{bmatrix} .$$

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Question 3

Find the inverse using the adjoint method:

$$(1) \begin{bmatrix} 5 & 2 \\ 1 & -1 \end{bmatrix}, \quad (2) \begin{bmatrix} 11 & -6 \\ 18 & -10 \end{bmatrix},$$

$$A^{-1} = \frac{\text{adj}(A)}{|A|} \rightarrow \text{cofactors}$$

$$(3) \begin{bmatrix} 1 & 0 & -2 \\ 4 & -2 & 1 \\ 1 & 2 & -10 \end{bmatrix}.$$

$$(1) A = \begin{pmatrix} 5 & 2 \\ 1 & -1 \end{pmatrix} \Rightarrow |A| = 5 \times (-1) - 2 \times 1 = -5 - 2 = -7$$

$$\text{adj}(A) = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{21} \\ c_{12} & c_{22} \end{pmatrix} = \begin{pmatrix} (-1)^{1+1} \times (-1) & (-1)^{1+2} \times 2 \\ (-1)^{2+1} \times 1 & (-1)^{2+2} \times 5 \end{pmatrix}$$

$$A^{-1} = \frac{\begin{pmatrix} -1 & -2 \\ -1 & 5 \end{pmatrix}}{-7} = \begin{pmatrix} \frac{1}{7} & \frac{2}{7} \\ \frac{1}{7} & -\frac{5}{7} \end{pmatrix} = \begin{pmatrix} -1 & -2 \\ -1 & 5 \end{pmatrix}$$

$$(3) A = \begin{pmatrix} 1 & 0 & -2 \\ 4 & -2 & 1 \\ 1 & 2 & -10 \end{pmatrix} \Rightarrow |A| = 1 \times (-2) \times (-10) + 0 + 4 \times 2 \times (-2) = -2$$

$$-1 \times (-2) \times (-2) - 0 - 1 \times 1 \times 2$$

$$\text{adj}(A) = \begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{21} & c_{31} \\ c_{12} & c_{22} & c_{32} \\ c_{13} & c_{23} & c_{33} \end{pmatrix} = \begin{pmatrix} (-1)^{1+1} \times \begin{vmatrix} -2 & 1 \\ 2 & -10 \end{vmatrix} & (-1)^{1+2} \times \begin{vmatrix} 0 & -2 \\ 2 & -10 \end{vmatrix} & (-1)^{1+3} \times \begin{vmatrix} 0 & -2 \\ -2 & 1 \end{vmatrix} \\ (-1)^{2+1} \times \begin{vmatrix} 4 & 1 \\ 1 & -10 \end{vmatrix} & (-1)^{2+2} \times \begin{vmatrix} 1 & -2 \\ 1 & -10 \end{vmatrix} & (-1)^{2+3} \times \begin{vmatrix} 1 & -2 \\ 4 & 1 \end{vmatrix} \\ (-1)^{3+1} \times \begin{vmatrix} 4 & -2 \\ 1 & 2 \end{vmatrix} & (-1)^{3+2} \times \begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix} & (-1)^{3+3} \times \begin{vmatrix} 1 & 0 \\ 4 & -2 \end{vmatrix} \end{pmatrix}$$

$$A^{-1} = \frac{\begin{pmatrix} 18 & -4 & -4 \\ 41 & -8 & -9 \\ 10 & -2 & -2 \end{pmatrix}}{-2}$$

$$= \begin{pmatrix} -9 & 2 & 2 \\ -\frac{41}{2} & 4 & \frac{9}{2} \\ -5 & 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 20 & -2 & -4 \\ -1 \times (40 - 1) & -10 + 2 & -1 \times (1 + 8) \\ 4 \times 2 + 2 & -1 \times 2 & -2 \end{pmatrix}$$

$$= \begin{pmatrix} 18 & -4 & -4 \\ 41 & -8 & -9 \\ 10 & -2 & -2 \end{pmatrix}$$

Tutorial Q3: Adjoint method I

Knowledge used

- Use

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj}(\mathbf{A}).$$

- For 2×2 matrices, compute determinant and cofactors directly.

Solution

For

$$\mathbf{A} = \begin{bmatrix} 5 & 2 \\ 1 & -1 \end{bmatrix}, \quad |\mathbf{A}| = -7.$$

$$\text{adj}(\mathbf{A}) = \begin{bmatrix} -1 & -2 \\ -1 & 5 \end{bmatrix}.$$

Hence

$$\mathbf{A}^{-1} = \begin{bmatrix} \frac{1}{7} & \frac{2}{7} \\ \frac{1}{7} & -\frac{5}{7} \end{bmatrix}.$$

Tutorial Q3: Adjoint method II

Knowledge used

- ▶ The same adjoint formula applies to any non-singular square matrix.
- ▶ Check $|\mathbf{A}| \neq 0$ before finding the inverse.

Solution

For

$$\mathbf{B} = \begin{bmatrix} 11 & -6 \\ 18 & -10 \end{bmatrix}, \quad |\mathbf{B}| = -2.$$

$$\text{adj}(\mathbf{B}) = \begin{bmatrix} -10 & 6 \\ -18 & 11 \end{bmatrix}.$$

Therefore,

$$\mathbf{B}^{-1} = \begin{bmatrix} 5 & -3 \\ 9 & -\frac{11}{2} \end{bmatrix}.$$

Tutorial Q3: Adjoint method III

Knowledge used

- For a 3×3 matrix, compute all cofactors.
- Then transpose the cofactor matrix to get the adjoint.

Solution

For

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & -2 \\ 4 & -2 & 1 \\ 1 & 2 & -10 \end{bmatrix}, \quad |\mathbf{A}| = -2.$$

The adjoint is

$$\text{adj}(\mathbf{A}) = \begin{bmatrix} 18 & -4 & -4 \\ 41 & -8 & -9 \\ 10 & -2 & -2 \end{bmatrix}.$$

Thus

$$\mathbf{A}^{-1} = \begin{bmatrix} -9 & 2 & 2 \\ -\frac{41}{2} & 4 & \frac{9}{2} \\ -5 & 1 & 1 \end{bmatrix}.$$

Tutorial Q4

Question 4

Solve the following system of equations by finding the inverse of the coefficient matrix using the adjoint method

6 min

$$2x_1 + 3x_2 - x_3 = -1,$$

$$5x_1 + x_2 + x_3 = -2,$$

$$-x_1 - x_2 + 4x_3 = 8.$$



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Question 4

Solve the following system of equations by finding the inverse of the coefficient matrix using the adjoint method

$$\textcircled{1} A^{-1} \begin{cases} \text{1/ } |A| \\ \text{2/ adj}(A) \end{cases}$$

$$\begin{cases} 2x_1 + 3x_2 - x_3 = -1, \\ 5x_1 + x_2 + x_3 = -2, \\ -x_1 - x_2 + 4x_3 = 8. \end{cases} \Leftrightarrow \begin{pmatrix} 2 & 3 & -1 \\ 5 & 1 & 1 \\ -1 & -1 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ 8 \end{pmatrix}$$

$$A \cdot X = b$$

$$\textcircled{2} x = ? \begin{cases} AX = b \\ A^{-1}AX = A^{-1}b \\ x = A^{-1}b \end{cases}$$

$$|A| = \begin{vmatrix} 2 & 3 & -1 \\ 5 & 1 & 1 \\ -1 & -1 & 4 \end{vmatrix} = -49$$

$$\text{adj}(A) = \begin{pmatrix} c_{11} & c_{21} & c_{31} \\ c_{12} & c_{22} & c_{32} \\ c_{13} & c_{23} & c_{33} \end{pmatrix} = \begin{pmatrix} (-1)^{1+1} \begin{vmatrix} 1 & 1 \\ -1 & 4 \end{vmatrix} & (-1)^{1+2} \begin{vmatrix} 3 & -1 \\ -1 & 4 \end{vmatrix} & (-1)^{1+3} \begin{vmatrix} 2 & -1 \\ 5 & 1 \end{vmatrix} \\ (-1)^{2+1} \begin{vmatrix} 2 & -1 \\ -1 & 4 \end{vmatrix} & (-1)^{2+2} \begin{vmatrix} 2 & -1 \\ -1 & 4 \end{vmatrix} & (-1)^{2+3} \begin{vmatrix} 2 & 3 \\ 5 & 1 \end{vmatrix} \\ (-1)^{3+1} \begin{vmatrix} 2 & 3 \\ 5 & 1 \end{vmatrix} & (-1)^{3+2} \begin{vmatrix} 2 & 3 \\ -1 & -1 \end{vmatrix} & (-1)^{3+3} \begin{vmatrix} 2 & 3 \\ 5 & 1 \end{vmatrix} \end{pmatrix}$$

$$A^{-1} = \frac{\begin{pmatrix} 5 & -11 & 4 \\ -21 & 7 & -7 \\ -4 & -1 & -13 \end{pmatrix}}{-49}$$

$$= \begin{pmatrix} 4+1 & -1 \times (12-1) & 3+1 \\ (-1) \times (20+1) & 8-1 & -1 \times (2+5) \\ -5+1 & (-1) \times (-2+3) & 2-15 \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{5}{49} & \frac{11}{49} & -\frac{4}{49} \\ \frac{3}{7} & -\frac{1}{7} & \frac{1}{7} \\ \frac{4}{49} & \frac{1}{49} & \frac{13}{49} \end{pmatrix}$$

$$= \begin{pmatrix} 5 & -11 & 4 \\ -21 & 7 & -7 \\ -4 & -1 & -13 \end{pmatrix}$$

$$x = A^{-1} \cdot b = \begin{pmatrix} -\frac{5}{49} & \frac{11}{49} & -\frac{4}{49} \\ \frac{3}{7} & -\frac{1}{7} & \frac{1}{7} \\ \frac{4}{49} & \frac{1}{49} & \frac{13}{49} \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -2 \\ 8 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{5}{49} - \frac{22}{49} - \frac{32}{49} \\ -\frac{3}{7} + \frac{2}{7} - \frac{8}{7} \\ -\frac{4}{49} - \frac{2}{49} - \frac{8 \times 13}{49} \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$$

Tutorial Q4: Inverse method

Knowledge used

- Write the system as $\mathbf{Ax} = \mathbf{b}$.
- If $\mathbf{A}^{-1}$ exists, then $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.

Solution

$$\mathbf{A} = \begin{bmatrix} 2 & 3 & -1 \\ 5 & 1 & 1 \\ -1 & -1 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -1 \\ -2 \\ 8 \end{bmatrix}.$$

$$\mathbf{A}^{-1} = \begin{bmatrix} -\frac{5}{49} & \frac{11}{49} & -\frac{4}{49} \\ \frac{3}{7} & -\frac{1}{7} & \frac{1}{7} \\ \frac{4}{49} & \frac{1}{49} & \frac{13}{49} \end{bmatrix}.$$

Hence

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}.$$

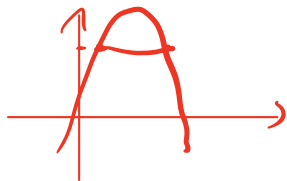
Therefore,

$$(x_1, x_2, x_3) = (-1, 1, 2).$$

Key takeaways

- ▶ Cramer's Rule solves variables using ratios of determinants.
- ▶ The adjoint method gives $\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj}(\mathbf{A})$ when $|\mathbf{A}| \neq 0$.
- ▶ Inverse matrices solve systems through $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.

A Few Final Words



$$\underbrace{3 \times 2} \cdot \underbrace{3 \times 2}$$

$$\underbrace{3 \times 2} \cdot 3 \times 3$$

- ▶ Thank you all for your company and support throughout this semester.
- ▶ Next semester, I will most likely be tutoring ECON2050 Mathematical Economics. Feel free to enrol if you are interested.
- ▶ Wish you all the best in your future studies.
- ▶ Hope you continue to keep your passion for learning and your curiosity alive.